

Homework 2

STAT 300 (Fall 2024)

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Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx_Yourfirstname.nb.html> file by the deadline.

- (1). Consider the fitted regression equation $\hat{y} = 100 + 15x$. Answer below
 - (a). What is the value of the sample slope?
 - (b). What is the predicted response when $x=0$?
 - (c). By how much the response is expected to increase for a unit increment of x ?
 - (d). If x is increased by 10 units by how much would you expect the response to increase?
- (2). For which of the following regression assumption(s), cannot be verified from a residual plot?
 - (a). Linearity (b). Constant variance (c). Independence (d). Zero mean
- (3). The number of calories and the grams of sugar per serving is measured for 36 breakfast cereal. The data is in *cereal.csv* file. We are interested in predicting calories using the sugar content.
 - (a). Make a scatterplot and comment on your observations?
 - (b). Find the least squares regression line for predicting calories based on the sugar content. Include your output and then write down (type) the least squares regression line
 - (c). Make a scatterplot and overlay the fitted regression model.
 - (d). Interpret the slope in the context of this problem
 - (e). Interpret the intercept in the context of this problem.
 - (f). Cheerios has 1 gram of sugar with 110 calories. Find the residual for this observation.
 - (g). Does the linear regression model appears to be a good summary of the relationship between calories and sugar content of breakfast cereal? Use appropriate graphs to justify your answer.
- (4). The data *playbill.csv* provides weekly reports on the box office ticket sales for plays on Broadway in New York. Provided data is for the week October 11–17, 2004 (referred to below as the current week). The data are in the form of the gross box office results for the current week and the gross box office results for the previous week (i.e., October 3–10, 2004).
 - (a) What would be the response and the predictor variable in this data?
 - (b) Make a scatter plot of the data to display the relationship between the two variables. Include the plot and comment on what you can see.
 - (c) Use R to obtain the least squares regression line for predicting this weeks' gross box office results from the previous week. Attach any R outputs.

- (d) Write the equation of the least squares regression line.
- (e) Make another scatterplot of the data and overlay the fitted regression line on the scatterplot. Use your favorite colour for the fitted line.

Here is a link to the R colour palette. <http://www.stat.columbia.edu/~tzheng/files/Rcolor.pdf>

(5). The table compares GDP (Gross Domestic Products) per Capita to Life Satisfaction; definitions of both are below. The goal is to explore the relationship between money and happiness Life Satisfaction scaled from 0 to 10 with zero being the least and 10 being best satisfied. GDP per Capita for 2015 Provided by International Monetary Fund. The gross domestic product of a nation divided by population of that nation.

Data source: *Kaggle.com*

- (a) What is the response and the explanatory variable in this setting?
- (b) Make a scatter plot of the data and comment on the relationship.
- (c) Calculate the sample correlation between the two variables?
- (d) Regardless your opinion in part (b) fit a simple linear regression model to this data. Show the R output of the SLR fit and the scatterplot with fitted line overlaid.
- (e) What is the coefficient of determination for the model. Interpret this value in the context of the problem.
- (f) In the original data, observation number 21 (“Luxembourg”) seems to be an outlier. Use the below code line to remove that observation from the dataset. Then re-fit the regression model to the reduced dataset. Include the R output and the scatterplot of the reduced data with fitted model overlaid.

```
#I named the original dataset "happy" and
#below line removed the observation number 21 and save it as a new data frame named "newhappy"
happynew<-happy[-21,]
```

- (g) Does it seem like model fit has improved once the “Luxembourg” was removed from the dataset. Explain.